SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

ASSESSMENT

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool	Code along exercise: Make your first HTML page together with CSS
Weightage	10%
Total Marks	10
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Scenario: Web Development Standards Implementation

A start-up is developing a modern web application and must address HTML elements, HTTP methods, XML syntax, and web communication protocols.

ANSWERS

Scenario: Web Development Standards Implementation

The following answers correspond to the five questions in Assessment Tool 3.

1. Select the correct HTML tag for video embedding and justify your answer.

Correct option: B — **<video src="video.mp4" controls></video>**
The HTML5 **<video>** element is designed to embed video content in a webpage. The **src** attribute specifies the video file, while **controls** displays built-in playback controls such as play, pause, and volume. The other options (**<media>**, **<movie>**, and **<vid>**) are not standard HTML elements for embedding video.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct option: B — **POST**.
POST sends the form data in the HTTP request body instead of placing it in the URL query string. It is generally appropriate for registration forms that submit user information. The application should use **HTTPS** to protect data while it travels between the browser and server, and the server must validate the submitted information. POST alone does not encrypt data.

3. Choose the proper semantic element for navigation and justify its usage.

Correct option: B — **<nav>**.
The HTML5 **<nav>** element identifies a section containing major navigation links, such as Home, About, Services, and Contact. It gives the page meaningful structure and helps browsers, assistive technologies, and developers identify the navigation area. A ** ** is generic, **<section>** is for a thematic grouping, and **<header>** represents introductory or heading content.

4. Identify the correct XML syntax and explain the error in other options.

Correct option: B — **<note><to>User</to><from>Admin</from></note>**.
This option is well-formed XML because every element has a matching closing tag and the child elements are correctly nested inside one root element. Option A is missing the closing **</note>** tag. Option C incorrectly closes **<to>** with **</from>**, so the tags do not match. Option D is incomplete because the opening **<note>** tag is not properly closed with a final **>**, and the element has no explicit closing tag.

5. Choose the correct protocol for web communication and justify your choice.

Correct option: B — **HTTP**.
HTTP (Hypertext Transfer Protocol) is used for communication between web clients, such as browsers, and web servers. It defines how requests and responses are exchanged to retrieve webpages and other web resources. In production, **HTTPS** is preferred because it uses TLS to protect communication. FTP is used for file transfer, SMTP for sending email, and SNMP for network management.

Code of Conduct

I, Naveen.S (Register Number: 192324248), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Student Signature: _____